Lesson 5.4 - Conditional Probability with Venn Diagrams and Contingency Tables

> **Video:** Conditional Probability With Venn Diagrams & Contingency Tables by The Organic Chemistry Tutor **Chapter:** 5 - Introduction to Probability

What it means

$P(A \mid B)$ is the probability that A happens **given that B has already happened**.

Formula:

$$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$$

Shortcut: ask "*how much of B is also A?*"

$$P(A \mid B) = \frac{\text{count in both } A \text{ and } B}{\text{count in } B}$$

Once B is known to have happened, B becomes the new sample space.

Example 1 - a die

$S = \{1, \ldots, 6\}$, A = odd = $\{1, 3, 5\}$, $B = \{3, 4, 5\}$

Shortcut: of the 3 outcomes in B, two (3 and 5) are in A:

$$P(A \mid B) = \frac{n(A \cap B)}{n(B)} = \frac{2}{3}$$

Formula:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{2/6}{3/6} = \frac{2}{3} \checkmark$$

Example 2 - Venn diagram (algebra and chemistry)

A school has **500** students. **150** take algebra (A), **80** take chemistry (C), and **30** take both.

The four regions of the Venn diagram (two overlapping circles inside a box of 500):

Region	Count
Algebra only (A but not C)	150 − 30 = **120**
Both ($A \cap C$, the overlap)	**30**
Chemistry only (C but not A)	80 − 30 = **50**
Neither (outside both circles)	500 − 200 = **300**
Total	**500**

(a) P(taking algebra)

$$P(A) = \frac{n(A)}{n(S)} = \frac{150}{500} = \frac{3}{10} = 0.30$$

(b) P(chemistry | algebra): "how much of A is also C?"

$$P(C \mid A) = \frac{n(C \cap A)}{n(A)} = \frac{30}{150} = \frac{1}{5} = 0.20$$

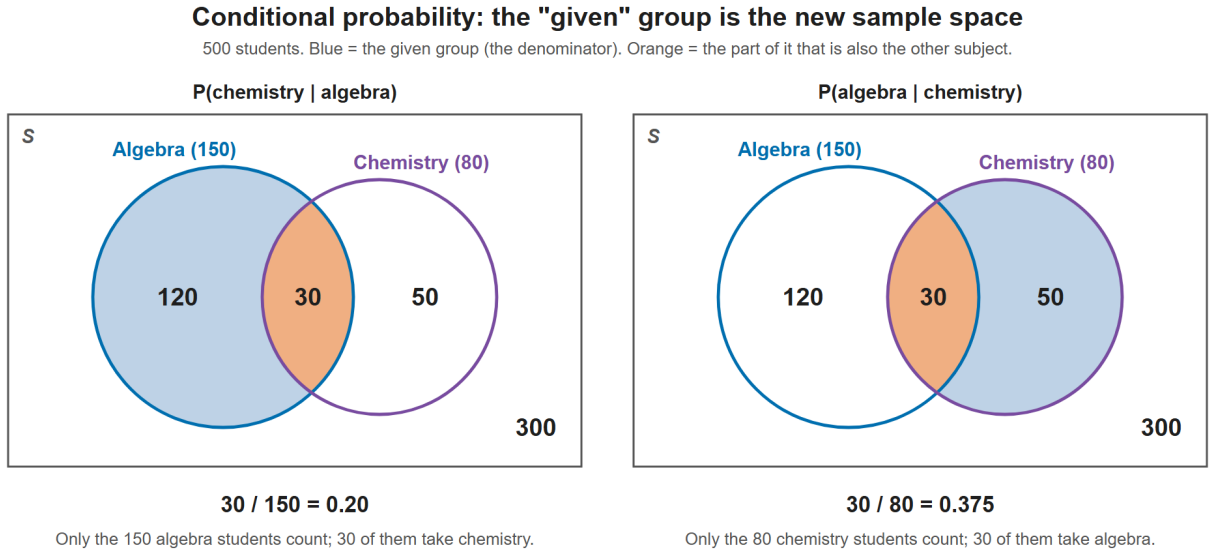
20% of the 150 algebra students also take chemistry: $0.20 \times 150 = 30$ ✓.

Formula check: $\frac{P(C \cap A)}{P(A)} = \frac{30/500}{150/500} = \frac{30}{150}$ ✓

(c) P(algebra | chemistry): "how much of C is also A?"

$$P(A \mid C) = \frac{n(A \cap C)}{n(C)} = \frac{30}{80} = \frac{3}{8} = 0.375$$

> $P(C \mid A) \neq P(A \mid C)$. The order matters, because the denominator is the "given" group.



The given group shrinks the sample space. P(chemistry | algebra) only looks at the 150 algebra students, and P(algebra | chemistry) only at the 80 chemistry students, so the same overlap of 30 gives two different answers.

Example 3 - contingency table (birds in a zoo)

200 birds: 70 male with brown eyes, 100 female with brown eyes, 20 male with blue eyes, 10 female with blue eyes.

	Blue eyes	Brown eyes	**Total**
Male	20	70	**90**
Female	10	100	**110**
Total	**30**	**170**	**200**

(a) P(female)

$$P(F) = \frac{\text{females}}{\text{total}} = \frac{110}{200} = \frac{11}{20} = 0.55$$

(b) P(male *and* brown eyes): a joint probability, not a conditional one

$$P(M \text{ and } Br) = \frac{\text{brown-eyed males}}{\text{total}} = \frac{70}{200} = 0.35$$

(c) P(female | brown eyes): restrict to the 170 brown-eyed birds

$$P(F \mid Br) = \frac{n(F \text{ and } Br)}{n(Br)} = \frac{100}{170} = \frac{10}{17} \approx 0.588$$

So 58.8% of the brown-eyed birds are female.

(d) P(male | blue eyes): restrict to the 30 blue-eyed birds

$$P(M \mid Bl) = \frac{n(M \text{ and } Bl)}{n(Bl)} = \frac{20}{30} = \frac{2}{3} \approx 0.667$$

(e) P(blue eyes | female): restrict to the 110 female birds

$$P(Bl \mid F) = \frac{n(Bl \text{ and } F)}{n(F)} = \frac{10}{110} = \frac{1}{11} \approx 0.091 = 9.1\%$$

> **Correction:** the video says 99.1% when converting this answer. $1/11 \approx 0.091$, which is **9.1%**.

Joint vs conditional probability in a contingency table

Orange = numerator. Blue outline = denominator. Light blue = the row or column you restrict to.

(b) P(male and brown)

	Blue	Brown	Total
Male	20	70	**90**
Female	10	100	**110**
Total	**30**	**170**	**200**

70 / 200 = 0.35

joint: divide by the grand total

(c) P(female | brown)

	Blue	Brown	Total
Male	20	70	**90**
Female	10	100	**110**
Total	**30**	**170**	**200**

100 / 170 ≈ 0.588

given brown: divide by the column total

(e) P(blue | female)

	Blue	Brown	Total
Male	20	70	**90**
Female	10	100	**110**
Total	**30**	**170**	**200**

10 / 110 ≈ 0.091

given female: divide by the row total

The numerator is always one cell. What changes is the denominator: the grand total for a joint probability, or one row or column total for a conditional probability.

Key takeaways

- $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$, or with counts, $\frac{n(A \cap B)}{n(B)}$.
- The "given" event becomes the **denominator**, the new reduced sample space.
- In a contingency table, a conditional probability uses **one row or column total** as the denominator. A joint probability uses the **grand total**.
- $P(A \mid B)$ and $P(B \mid A)$ are usually different.